PIEFS: Physics-Informed Eigenfunction Features with Learnable Scaling

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Background & Motivation

Core Challenge: Spectral methods extract robust geometric and manifold features but typically rely on fixed kernels, rigid graph Laplacians, or manually tuned feature scaling.

Limitations of Classical Spectral Methods:

- Features map directly to fixed, supervision-independent operator eigenfunctions
- Eigencomputation costs scale poorly with dataset size ($O(N^3)$ matrix diagonalization)
- Test-time evaluation on out-of-sample data points forces an expensive rebuild of the graph and its entire spectrum

Our Contribution: We introduce **PIEFS**, a supervised representation learning framework with a spectral inductive bias. By parameterizing continuous coordinates with neural networks, it produces task-adaptive, Dirichlet-regularized features with cheap inference.

Core Idea

Instead of minimizing a single Rayleigh quotient on a discrete mesh, PIEFS parameterizes K continuous scalar neural coordinate functions $\varphi_1(x), \varphi_2(x), \dots, \varphi_K(x)$ with disjoint θ_j parameters optimized via sequential block-coordinate descent

The Modified Dirichlet Energy (MDE)

To embed geometric awareness, we establish a continuous bilinear form weighted by the data density $p(x)$

$$\mathcal{L}_{mde} = \int \|A(x)\nabla\phi_k(x)\|^2 p(x)dx$$

where A(x) is a jointly learned data-dependent metric tensor. Rather than smoothing features isotropically across the whole domain, A(x) alters the regularizing field based on task dynamics.

Matrix Parametrization & Stability

To prevent optimization collapses, the metric tensor undergoes a volume-preserving factorization:

$$A(x) = \Lambda(x)U(x)$$

- **Isotropic Baseline (OFF)**
$A(x) = I$, yielding standard Euclidean Dirichlet regularization
- **Anisotropic Scaling (DIAG)**
$A(x) = \Lambda(x)$, where log-eigenvalues are constrained by a zero-mean MLP head such that $\det\Lambda(x) = 1$
- **Structured Rotation (TROTTER)**
To avoid expensive runtime QR or Gram-Schmidt loops, the orthogonal factor $U(x) \in SO(d)$ is parameterized via a structured product of d-1 Givens plane rotations:

$$U(x) = R_{d-2}(\omega_{d-2}) \cdots R_1(\omega_1)\, R_0(\omega_0)$$

Angles are strictly bounded via $\omega_i(x) = \pi \tanh(\mathrm{MLP}_i(x))$ to ensure numerical stability and eliminate unbounded extrapolation.

Key Experimental Results

Supervised Downstream Performance

PIEFS shows strong convergence within 15,000 steps across diverse geometries

Downstream accuracy (%, mean±std, 5 seeds). Bold = best PIEFS variant per row

Dataset	RF	PCA + LR	NeuralEF	PIEFS (OFF)	PIEFS (TROTTER)
Two Moons	99.89±0.04	88.15±0.22	88.23±0.31	99.87±0.00	99.88±0.03
Circles	99.96±0.04	50.67±5.12	100.00±0.00	99.99±0.03	100.00±0.00
MNIST (K=16)	97.07±0.03	92.20±0.00	82.52±0.29	95.41±0.32	94.94±0.20
CIFAR-10 (K=16)	82.65±0.11	86.82±0.00	76.05±0.55	85.50±0.53	85.22±0.39

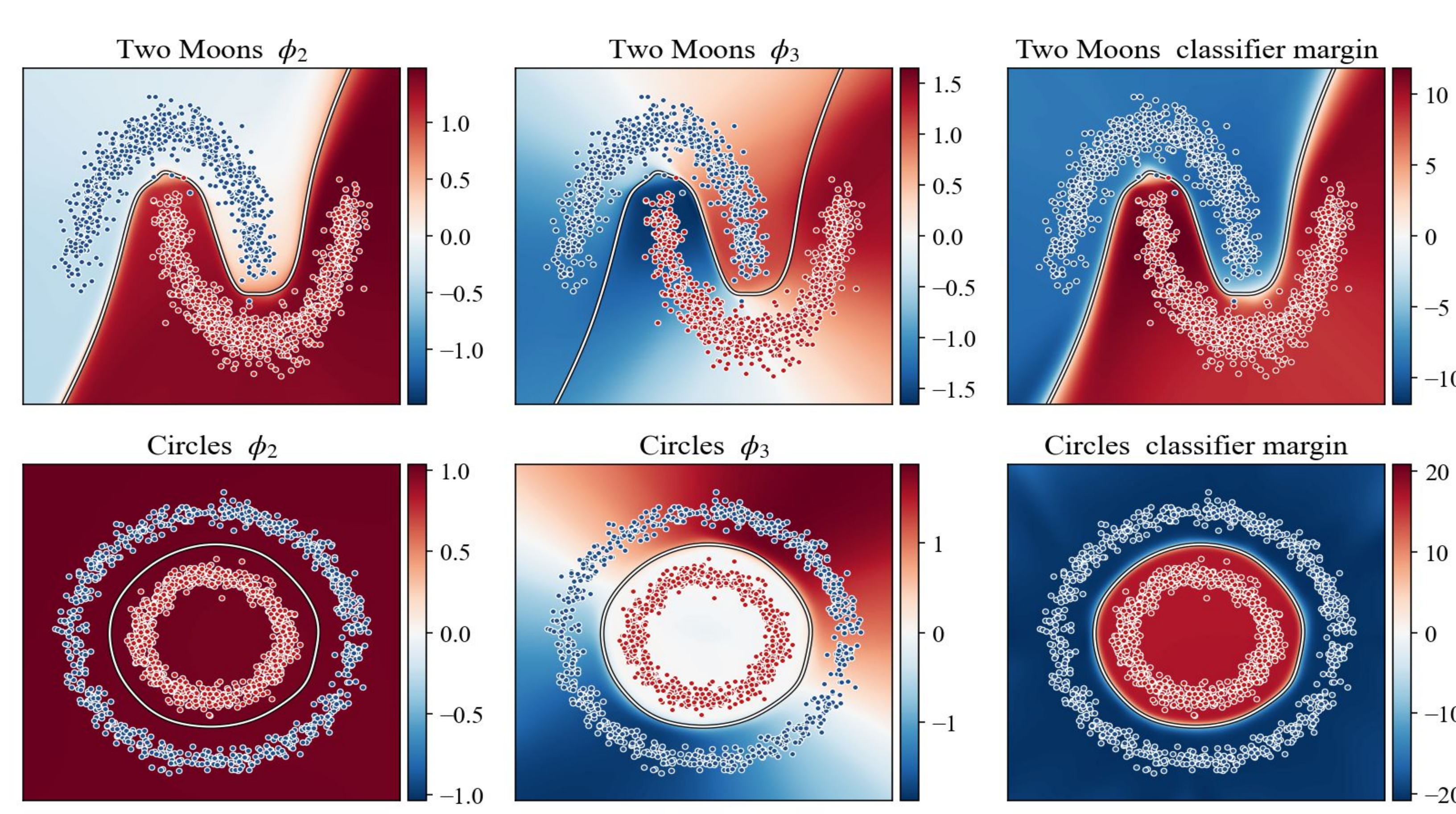
Loss Function

PIEFS balances three goals per coordinate:

- **Orthogonality** — Gram matrix ≈ identity → distinct modes.
- **Supervision** — cross-entropy on a linear readout → task-adaptive features.
- **Smoothness (MDE)** — Dirichlet penalty regularizes along data geometry.

A dynamic weighting scheme sets the priority each step: orthogonality → classification → smoothness. Coordinates are trained sequentially (one map at a time, earlier ones frozen), keeping inference cheap.

Classifier Geometry (PIEFS-off) and Functions Visualisation



Gram-vs-accuracy (Supervised / Unsupervised)